

WORK AND ENERGY

Definitions

Energy: a measure of a body's (or system's) ability to do work.

Work: is done when an applied force causes an object to move in the direction of the force.

The work done by a force on an object is given by:

$$W = Fs$$

where W = the work done by the force (Joules - J).
 F = the applied force (Newtons - N).
 s = the displacement of the object in the direction of the applied force (metres - m).

Examples

1. A box of mass 15.0 kg has a horizontal force of 2.00×10^2 N applied to it and it moves 4.50 m across a floor. What work has been done on the box?

$$\begin{aligned} W &= Fs \\ &= (2.00 \times 10^2)(4.50) \\ &= 9.00 \times 10^2 \text{ J} \end{aligned}$$

$$\therefore \text{Work done} = 9.00 \times 10^2 \text{ J}$$

2. A car moving at 60.0 kmh^{-1} due south has its brakes applied and it uniformly stops in 35.0 m. If the work done by the brakes is 8.87×10^4 J, calculate the average force exerted by the brakes.

$$\begin{aligned} W &= Fs \\ \Rightarrow F &= \frac{W}{s} \\ &= \frac{8.87 \times 10^4}{35.0} \\ &= 2.534 \times 10^3 \text{ N} \end{aligned}$$

$$\therefore \text{Average force exerted} = 2.53 \times 10^3 \text{ N north}$$

THE CAR SLOWS DOWN SO THE APPLIED FORCE MUST BE IN THE **OPPOSITE** DIRECTION TO ITS MOVEMENT .

FORCE IS A VECTOR AND NEEDS A DIRECTION GIVEN.

POTENTIAL AND KINETIC ENERGY

Potential Energy (E_p)

This is the energy a body has due to the arrangement of its particles or due to its position above some level.

Something needs to happen to the system to release this potential energy.

Examples

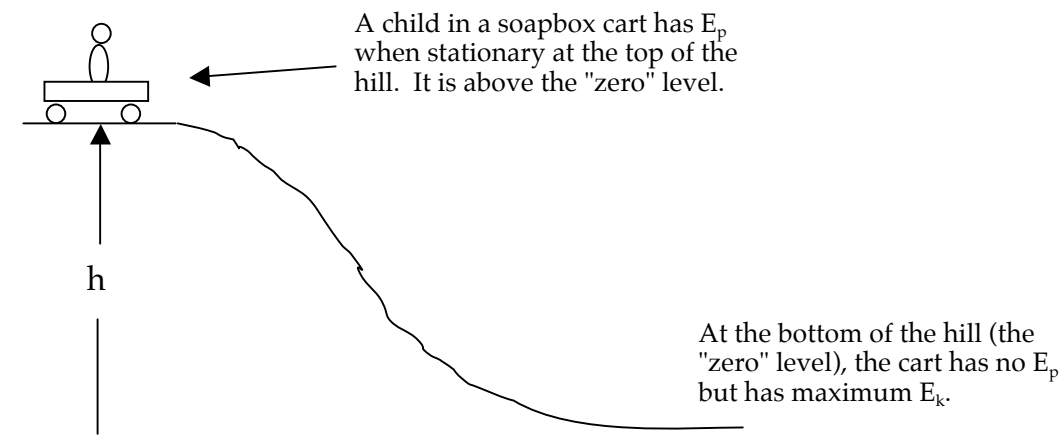
Petrol - has stored E_p in its chemical bonds that is released as energy if it is burned.

An object at a height - they can potentially be pulled by gravity to the lowest level.

Gravitational Potential Energy

This is the energy possessed by a body due to its height above a given level.

Any object above a given "zero" level has the ability to release this energy as E_k as it falls to this "zero" level.



The amount of potential energy is given by:

$$E_p = mgh$$

where

- E_p = the potential energy of the body (J)
- m = the mass of the body (kg)
- g = the acceleration due to gravity (ms^{-2})
- h = the height above the "zero" level (m)

Example

A 2.52 kg clock is sitting on a table, 1.52 m above the ground. Calculate the potential energy of the clock.

$$\begin{aligned}E_p &= mgh \\&= (2.53)(9.80)(1.52) \\&= 37.69 \text{ J}\end{aligned}$$

$$\therefore \underline{E_p = 37.7 \text{ J}}$$

Kinetic Energy (E_k)

This is the energy due to a body's movement.

The kinetic energy of a body due to its movement is given by the equation:

$$\boxed{E_k = \frac{1}{2} mv^2}$$

where : E_k = kinetic energy due to movement (J)
 m = mass of the object (kg)
 v = velocity of the body (ms^{-1})

Example

A boy is riding a bike at 12.0 ms^{-1} . If the bike and boy combine weigh 67.8 kg, calculate the kinetic energy of the system.

$$\begin{aligned}E_k &= \frac{1}{2} mv^2 \\&= \frac{1}{2} (67.8)(12.0)^2 \\&= 4.882 \times 10^3 \text{ J}\end{aligned}$$

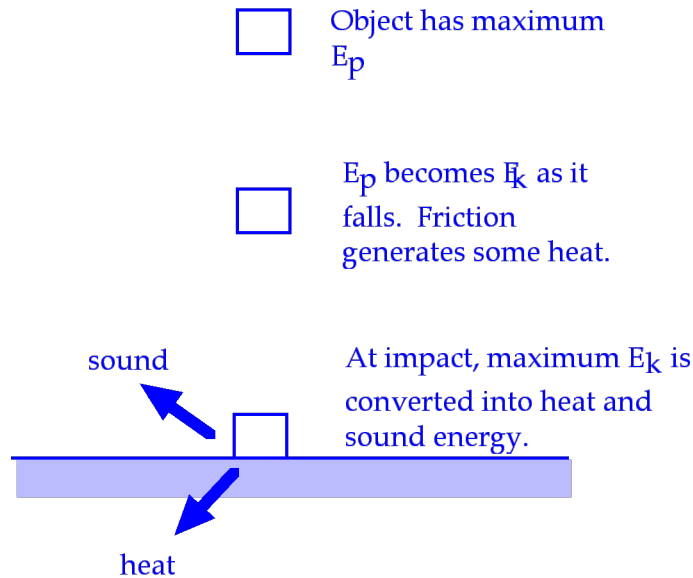
$$\therefore \underline{E_k = 4.88 \times 10^3 \text{ J}}$$

CONSERVATION OF MECHANICAL ENERGY

In all situations, energy must be conserved - it can be transformed from one form to another, but the total amount stays the same.

Example

Consider an object dropped from a height onto a hard surface.



The above situation describes a particular case where the total mechanical energy is conserved.

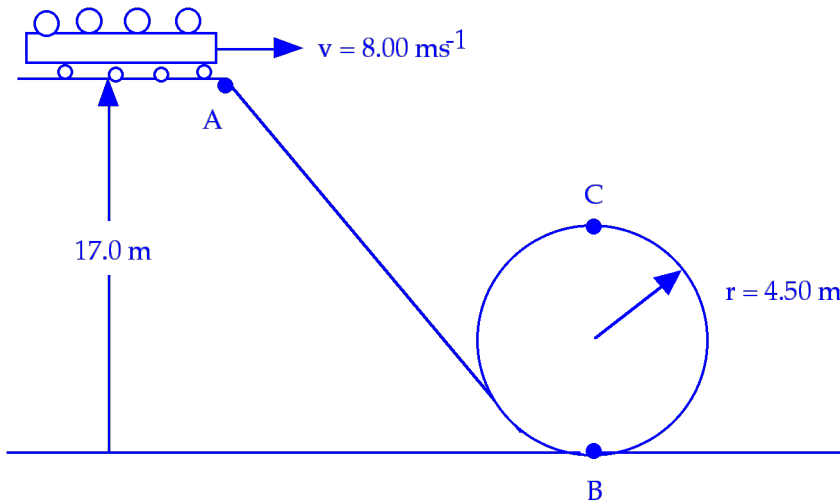
$$\begin{aligned} E_{\text{total}} &= E_p + E_k \\ &= mgh + \frac{1}{2}mv^2 \end{aligned}$$

NOTE

In calculations, it will generally be considered that friction is not acting.

Example

A roller coaster of mass $3.70 \times 10^3 \text{ kg}$ is moving at 8.00 ms^{-1} at point A as shown below.



- (a) Determine the total energy of the coaster at A.
- (b) Calculate the speed achieved at point B.
- (c) Does the coaster safely make it through the loop if the minimum speed to do so is 6.64 ms^{-1} ? Justify your answer by calculation.

- (a) At point A:

$$\begin{aligned} E_t &= E_p + E_k \\ &= mgh + \frac{1}{2}mv^2 \\ &= (3.70 \times 10^3)(9.80)(17.0) + \frac{1}{2}(3.70 \times 10^3)(8.00)^2 \\ &= 7.348 \times 10^5 \text{ J} \end{aligned}$$

$$\therefore \underline{E_t = 7.35 \times 10^5 \text{ J}}$$

- (b) At point B:

$$\begin{aligned} E_t &= E_k = \frac{1}{2}mv^2 \\ \Rightarrow 7.348 \times 10^5 &= \frac{1}{2}(3.70 \times 10^3)v^2 \\ \Rightarrow v &= 19.93 \text{ ms}^{-1} \end{aligned}$$

$$\therefore \underline{v = 19.9 \text{ ms}^{-1}}$$

(c) At point C:

$$\begin{aligned} E_t &= E_p + E_k \\ &= mgh + \frac{1}{2}mv^2 \end{aligned}$$

Remember to double
the radius.

$$\Rightarrow 7.348 \times 10^5 = (3.70 \times 10^3)(9.80)(9.00) + \frac{1}{2}(3.70 \times 10^3)v^2$$

$$\Rightarrow v = 14.86 \text{ ms}^{-1} > 6.64 \text{ ms}^{-1}$$

$\therefore$ The roller coaster makes it safely.